



Grantee

Urban Revival: Analyzing urbanization through changing land cover and its effects on surrounding communities

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Introduction

Remote sensing leverages advanced satellites and sensors to observe and analyze Earth's surface, playing a crucial role in environmental monitoring, urban planning, and climate studies. It encompasses two primary methods: passive remote sensing, which relies on detecting sunlight or heat naturally reflected or emitted by the surface (useful for studying vegetation, water bodies, and urban areas), and active remote sensing, such as radar and lidar, which sends its signals to measure Earth's features, even at night or through clouds. Prominent satellites like Sentinel-1 (radar), Sentinel-2 (multispectral), the Landsat series, and WorldView-4 provide diverse and complementary data. For example, Sentinel-1's radar detects surface changes under any weather, Sentinel-2 delivers detailed optical images across several wavelengths, Landsat satellites supply decades of historical observation for change detection, and WorldView-4 offers very high-resolution imagery suitable for urban mapping. By combining these satellite datasets with real-world, "ground-truth" observations, researchers and urban planners gain accurate and up-to-date land cover maps. This integration not only improves our understanding of how landscapes transform but also supports informed decisions for sustainable development, green space preservation, and environmental resilience in the face of rapid change.

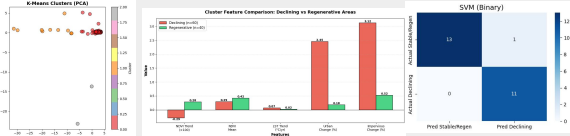


Abstract

Urban development is altering land cover globally, impacting ecosystems, biodiversity, and climates. Our study investigates how satellite imagery and remote sensing can identify patterns of urban development and analyze related environmental changes. We used initial field observations from the GLOBE Observer App, where citizen scientists collect high-resolution, ground-level, using their smartphone cameras. Each observation included geotagged photographs taken in standardized directions. This data revealed that existing land cover classifications varied in accuracy, prompting our investigation into their value and in future research. This led to our primary research question: How effectively can remote sensing detect urban expansion and its impacts on surrounding communities? Using urban cities like Chicago, Atlanta, Los Angeles, Detroit, and Austin, we studied impacts such as green space gains/losses, heat island formations, and nighttime brightness changes. To do this, we utilized Landsat and Sentinel-2 imagery, NDVI, ECOSTRESS, and VIIRS data. Integrating multiple remote sensing datasets, we trained a machine learning model that classified geographic zones into three categories of development: regenerative, unequal, and stable. This classification differentiated areas that experienced positive environmental and urban recovery, harmful development, and minimal change. Our findings demonstrate that remote sensing can reveal patterns of urban growth and their impact on communities. Moving forward, the zone classifications could be used as a valuable resource for urban planners and local communities. By identifying areas at risk of environmental degradation, intervention could be more targeted and effective. This framework could also be extended to other cities to support global comparisons of urban development.

Keywords: Urbanization, Satellite, Regeneration, Land Cover, Remote Sensing

Results



Two models demonstrated the strongest performance:

- a. Support Vector Machine (RBF kernel): 100% cross-validation accuracy and 96% test accuracy.
- b. Logistic Regression: 98–100% cross-validation accuracy and 96–100% test accuracy.

Both models perfectly separated the refined classes in the test set. Most influential predictors included NDVI mean, green decline index, impervious surface change, and NDVI-to-LST ratio. The refined cluster distribution showed that roughly 60% of study areas exhibited declining environmental conditions, while 40% showed stable or regenerative patterns.

Collected AOI Data

Platform	Landsat 5-9	WorldView-4	Sentinel-1/2				GLOBE Observer						Collect Earth Online
		WorldView-4	WorldView-4	Sentinel-2	Sentinel-2	Up	Down	west	south	east	north	Map Overview Map Information	
AOI	Landsat Time Series Graph												
1													
2													
3													
4													
5													
6													
7													
8													
9													
10													

After looking through the over 200 observed locations, we found some interesting trends on urbanization. For example, in Taylorsville, Utah, we found that the satellite data didn't account for newly built homes, and misclassified this area as grasslands. Additionally, in Superior, Colorado, we found through Landsat Time Series images that developed land increased by 150% between 2000 and 2025. Both of these locations highlighted the dangerously rapid urban development taking place, especially in suburbs. More generally, we found that fractional impervious surface increased in all locations over the past 20+ years. These findings led to our research question, **how effectively can remote sensing detect urban expansion and its impacts on surrounding communities?** Through answering this question, we seek to find solutions to the effects rapid urbanization has on communities, including but not limited to greenery loss, land surface temperature increase, and gentrification.

Methods

To address our research question, we created a four-step workflow.

Step 1: Study Area Selection
We selected five major U.S. cities—Chicago, Atlanta, Detroit, Austin, and Los Angeles—as representative case studies of diverse urban development patterns. Within each city, we identified 20 regions (100 total) for focused analysis. Initial labels (regenerative, unequal, stable) were assigned based on reviews of news articles, academic literature, urban development reports, and manual inspection of historical satellite imagery. These early labels provided a starting point but were intentionally treated as provisional, given the potential for bias or misinterpretation during manual classification.

Step 2: Satellite Data Extraction
We extracted a comprehensive set of satellite-based environmental indicators using the Google Earth Engine API. Our dataset included NDVI trends and mean values from Landsat 7 ETM+ (30 m), land surface temperature (LST) trends and means from MODIS Terra (1 km), MODIS MCD12Q1 land-cover transitions, and impervious surface change from the National Land Cover Database (NLCD). To capture long-term patterns rather than short-term variation, we calculated trend values across 5-year intervals from 2001–2020. This time-series approach allowed us to distinguish areas undergoing gradual environmental decline from those experiencing stable or regenerative conditions.

Step 3: Feature Engineering and Label Refinement
To address inconsistencies in the initial manual labels, we applied K-Means clustering (k=3) to reveal natural groupings based on vegetation, temperature, and urbanization trends. We then engineered additional features—such as NDVI-to-LST ratio, urbanization rate, green decline index, development pressure, and normalized changes—to better quantify interactions between vegetation, heat, and built environment expansion. Cluster patterns guided the creation of a balanced binary labeling scheme: Stable/Regenerating (~40%) and Declining/Changing (~60%).

Step 4: Machine Learning Classification
We evaluated several models, including Support Vector Machine (RBF), Logistic Regression, Random Forest, Gradient Boosting, XGBoost, and a simple Neural Network. Features were standardized using RobustScaler, and a stratified 75/25 train-test split with 5-fold cross-validation ensured reliable estimates. A balanced dataset was also created through undersampling to test model robustness, and performance was measured using accuracy, F1-score, and confusion matrices.

Conclusion

This project shows that combining satellite remote sensing with clustering and machine learning provides a highly accurate and scalable way to assess urban environmental change. Initial labels based on field observations and manual interpretation often failed to match real satellite patterns, but applying K-Means clustering revealed the true structure in the data. This data-driven relabeling, paired with engineered features, increased model performance from below 35% to near-perfect accuracy.

Our strongest models, especially SVM and Logistic Regression, clearly separated declining areas from stable or regenerating ones across all five cities. Key features such as the NDVI-to-LST ratio, green decline index, and urbanization rate effectively captured interactions between vegetation, heat, and built development. The refined labels showed that roughly 60% of regions exhibit signs of environmental decline.

The framework is flexible and can be expanded to additional cities or integrated with other datasets such as nighttime lights or demographic indicators. Because it relies entirely on satellite data and automated classification, it offers a practical tool for planners and researchers to monitor environmental conditions and identify areas where intervention is most needed.

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